RUPAN CHAKKARAVARTHY E

Technical Lead — Solutions · AI Platforms · XR & Real-Time Systems

Chennai, India | +91 88076 26297 | erupanc@gmail.com

github.com/RupanRC | Portfolio: rupanrc.vercel.app

PROFILE

Technical Lead — Solutions with ten years spanning two hard domains: real-time defense simulation (fighter-jet simulators, ARINC 661 cockpit displays, maritime situational awareness, XR hardware integration) and, since 2024, agentic AI platforms. Currently architecting AI-driven software delivery systems that orchestrate fleets of coding agents through spec, plan, implement, review and QA phases with human approval gates. Comfortable owning everything from GLSL shaders and Ada integration to RAG pipelines, NL-to-SQL engines and air-gapped Linux delivery. Team lead and technical manager for national aerospace and defense R&D programs.

CORE SKILLS

AI & Agents: Claude / Claude Code multi-agent orchestration, MCP, RAG (RAGFlow, Chroma), Vanna NL→SQL, Ollama local LLMs, code knowledge graphs, agent-native architecture

XR & Simulation: Unity 3D, C#, MRTK3, ARCore / ARFoundation, OpenVR / SteamVR, FlightGear, Varjo XR-3/4, Meta Quest, HTC Vive, Leap Motion, Xsens Movella, SenseGlove

Real-Time & Avionics: C, C++, OpenGL, ARINC 661, Ada embedded integration, Qt, ROS2 / NAV2, GLSL/HLSL shaders, OpenCV, UDP video streaming

Web & Platform: TypeScript, React, Vite, Node.js, Express, FastAPI, Tauri v2, SSE, SQLite / MySQL / MSSQL, Tailwind, auth (scrypt, HMAC, RBAC)

Ops & Delivery: Docker, nginx, air-gapped Ubuntu deployment, Git worktree orchestration, TDD, automated browser testing, WBS / SOW / costing

WORK EXPERIENCE

Technical Lead — Solutions — MicroGenesis TechSoft Pvt Ltd *Mar 2024 – Present*

- Deployed with the Aeronautical Development Agency (DRDO, Ministry of Defence) for indigenous XR real-time pilot ergonomics study, and with CAIR-DRDO as Technical Manager for GIS, real-time and robotics programs.
- Architected and solely built an AI-SDLC orchestration platform driving Claude coding agents through spec → plan → implement → review → QA with human gates: dependency-wave scheduler running 8 parallel agents in isolated git worktrees, automated browser verification of acceptance criteria, code-knowledge-graph context layer, local-model (Ollama) routing for on-premise use.
- Delivered an AR quality-inspection product for a manufacturing client : Unity + MRTK3 headset experience, ARCore tablet port at full parity verified on physical hardware with test-driven EditMode and E2E suites, plus a React/Express management portal with rework traceability and first-pass-yield analytics.
- Built a flight-data insights platform: RAG over raw .dat telemetry, NL→SQL (Vanna 2.0) verified live across 13 datasets, MPEG-2 video-evidence pipelines — packaged for fully air-gapped Ubuntu deployment (Docker, offline installers, patch bundles).
- Designed and shipped an offline↔online secure data-exchange PoC (ARINC-style signed, encrypted change-set packaging with field-level review).
- Led commercial engineering: tender analysis to costed proposals, WBS/SOW generation tooling, market research across 125 competing agent-orchestration projects, pitch decks and phased rebuild programs.

Freelance Software Engineer — Independent *Mar 2023 – Mar 2024*

- Developed interactive applications; ran day-to-day project coordination, planning and implementation; authored functional and technical documentation.

Senior Engineer — COMAVIA Systems Technologies Pvt Ltd *Nov 2021 – Feb 2023*

- Developed ARINC 661-compliant Cockpit Display System software for indigenous fighter programs (MK2, AMCA) in C, C++ and OpenGL — low-latency SVG-based display pages, shader-based rendering, multi-partition dynamic displays. Team lead.
- Integrated Epiphan video capture with real-time cropping; UDP video streaming via OpenCV; Ada ↔ C++ integration over DLLs and network protocols for airworthy test units.
- Developed IMSAS — Indian Maritime Situational Awareness System: GIS-based real-time common operating picture in Qt/C++ with PostgreSQL storage and Kafka / RabbitMQ event streaming across consoles.

Senior Software Engineer (XR) — BAeHAL Software Limited *Jan – Oct 2019, Nov 2020 – Oct 2021*

- Deployed with the Aeronautical Development Agency for the indigenous VR fighter-jet simulator: HUD shaders (world-space UV), physics-joint cockpit switches, Leap Motion hand tracking, editor-scripted auto-rigging of cockpit controls from engineering spreadsheets, multi-surface symbology scheduling. Team lead.

Senior XR Engineer / XR Engineer — D2VR Technologies *Jul – Dec 2018, Oct 2019 – Oct 2020*

- Built VR/AR applications and prototypes; agile planning, code reviews, risk identification and workflow improvement.

Unity 3D Developer — Satya Aerospace Tech Data Services Pvt Ltd *May 2016 – Jun 2018*

- Game programming, network gameplay code, and VR builds for Windows MR, Oculus Rift and HTC Vive; AR with Vuforia, ARCore, Google VR.

KEY PROJECTS

- **FORGE:** AI software delivery platform — agent orchestration with human gates, knowledge-graph context, autonomous defect-fix loop (an automated browser test caught a data-loss bug; the system repaired and re-verified itself).
- **Qualia:** AR quality inspection for manufacturing — headset + tablet parity, TDD throughout.
- **Flight Data Insights:** RAG + NL→SQL telemetry analysis, air-gapped delivery for flight-test environments.
- **Supplier Intelligence:** Component supply-chain risk intelligence — 187 systems risk-scored, market cross-referencing, disruption simulation; full costed delivery program (5,692 hours).
- **Defense simulation:** Indigenous VR fighter-jet simulator and ARINC 661 cockpit display systems for national fighter programs.
- **Pilot ergonomics:** Xsens motion-capture ergonomic analysis of pilots under high-load maneuvering, with automated reporting.
- **IMSAS:** Indian Maritime Situational Awareness System — GIS-based real-time common operating picture. Qt, C++, PostgreSQL, Kafka, RabbitMQ.
- **Simulation & CV:** Bomb-drop trajectory simulation (FlightGear/Unity), fighter-jet sensors simulation with multi-monitor XLIB windows on Ubuntu, synthetic-terrain shaders in FlightGear, lane-detection CV integration, UDP video streaming via OpenCV.
- **Avionics & robotics:** ECFM LRU test-case generation and component testing for an airworthy unit (Ada ↔ C++), AUV ECA-A9 software on ROS2/NAV2, multi-region large-area touch displays in Windows C++.
- **XR & interactive:** LIDAR point-cloud VR walkthroughs with MIT Chennai, museum interactives (Puli-Meka multiplayer AI strategy game, 4-way touch table, VR 360° tour), AR sketch app with SLAM anchoring, MMORPG VR building/terrain/networking systems.
- **Self-hosted AI infrastructure:** Multimodal LLM management backend and open-source chat platform with user/model access control (Docker), VS Code AI pair-programming rollout on local LLMs, low-latency self-hosted video conferencing.

EDUCATION

B.E. Computer Science — MNM Jain Engineering College *2012 – 2016*